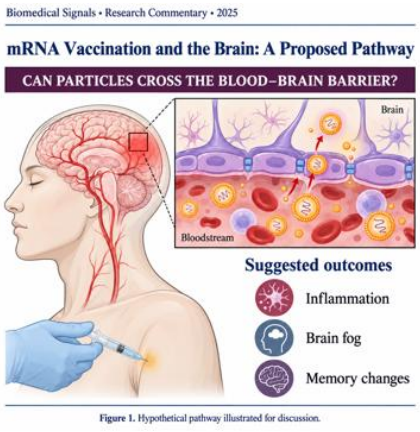
SDA Vision / Research report

journal-diagram.png



Assessment summary

Field	Recorded detail
Finding	Leaning synthetic
Recorded verdict	synthetic likely
Confidence	medium
Indicative rating	68/100
Rating basis	median of available model ratings; not a calibrated probability
Explanation	2 models agree this is likely synthetic.
Coverage	image - 1 frame(s) analysed
Review context	Research assessment. Review alongside source information and context. When credentials are missing, origin remains unresolved by this check. Use these findings to inform a documented human review.

Provider overview

Provider	Type / status	Assessment	Rating
Claude Vision	vision / ok	synthetic likely	80/100
OpenAI Vision	vision / ok	inconclusive	45/100
Gemini Vision	vision / ok	synthetic likely	68/100
C2PA Content Credentials	provenance / ok	inconclusive	n/a
Local forensic cues	forensic / ok	partially synthetic	41/100

Provider evidence / 1

Field	Recorded detail
Provider	Claude Vision
Model	claude-opus-5-5
Status	ok
Assessment	synthetic likely
Confidence	medium
Role in assessment	Supporting
Provider summary	synthetic likelihood 80/100; tells: uniform soft airbrushed painterly shading across all elements, typical of current image-generation models, whole composite (portrait, anatomy, inset, icons, typography) rendered in one seamless style rather than assembled from separate assets, anatomically loose vasculature: arteries drawn over the brain surface and an idealized carotid path from neck to cortex, generic glowing highlights (red cortical hotspot, yellow injection-site glow) used as stylized emphasis
Evidence 1	Uniform soft airbrushed painterly shading across all elements, typical of current image-generation models
Evidence 2	Whole composite (portrait, anatomy, inset, icons, typography) rendered in one seamless style rather than assembled from separate assets
Evidence 3	Anatomically loose vasculature: arteries drawn over the brain surface and an idealized carotid path from neck to cortex
Evidence 4	Generic glowing highlights (red cortical hotspot, yellow injection-site glow) used as stylized emphasis
Evidence 5	Repetitive near-identical lipid-nanoparticle and red blood cell motifs with slight random variation
Evidence 6	Clean but slightly generic serif typography and masthead ('Biomedical Signals'), consistent with high-fidelity text rendering in recent generators
Evidence 7	No camera-origin signals (no sensor noise, lens blur, or optical depth of field); content is digital illustration throughout
Evidence 8	Gloved hand and syringe are plausible but smoothly idealized, with simplified plunger and needle detail
Evidence 9	Caveat: This is an illustration, not a photo, so camera-forensic cues do not apply. Skilled human illustrators using digital airbrush styles can produce similar output. Check for C2PA/metadata, reverse-image search, and verify whether 'Biomedical Signals' is a real publication.

Provider evidence / 2

Field	Recorded detail
Provider	OpenAI Vision
Model	gpt-6-astra
Status	ok
Assessment	inconclusive
Confidence	low
Role in assessment	Additional finding
Provider summary	synthetic likelihood 45/100; tells: Entire image is a medical illustration, with no camera-photo base visible., Headings and labels are legible, with consistent serif typography and aligned layout., Anatomy and cellular elements use smooth shaded fills and clean outlines compatible with conventional digital illustration., Repeated particle motifs have variable internal squiggles, but no decisive generation artifact is visible.
Evidence 1	Entire image is a medical illustration, with no camera-photo base visible.
Evidence 2	Headings and labels are legible, with consistent serif typography and aligned layout.
Evidence 3	Anatomy and cellular elements use smooth shaded fills and clean outlines compatible with conventional digital illustration.
Evidence 4	Repeated particle motifs have variable internal squiggles, but no decisive generation artifact is visible.
Evidence 5	Caveat: Conventional digital artwork and AI-assisted illustration can look alike; source files and provenance are needed to distinguish them.

Provider evidence / 3

Field	Recorded detail
Provider	Gemini Vision
Model	gemini-3.8-flash
Status	ok
Assessment	synthetic likely
Confidence	medium
Role in assessment	Supporting
Provider summary	synthetic likelihood 68/100; tells: model read this as AI editing over a photographic base image, sterile digital rendering style, slightly inconsistent anatomical blending in syringe and hand, typical AI-generated vector/digital medical illustration texture
Evidence 1	Model read this as AI editing over a photographic base image
Evidence 2	Sterile digital rendering style
Evidence 3	Slightly inconsistent anatomical blending in syringe and hand
Evidence 4	Typical AI-generated vector/digital medical illustration texture
Evidence 5	Clean modern typography composited over generated artwork
Evidence 6	Caveat: High-quality human digital medical illustrators use similar styles, making definitive technical separation difficult without source files.

Content Credentials / provenance

Field	Recorded detail
Provider	C2PA Content Credentials
Model	Not recorded
Status	ok
Assessment	inconclusive
Confidence	Not recorded
Role in assessment	Additional finding
Provider summary	Embedded credentials: none found. Origin remains unresolved by this check.
Evidence 1	Coverage: credentials embedded in the submitted file. External credentials were not retrieved.
Evidence 2	Verifier: c2pa-python 0.37.12; native SDK 0.91.0; specification target 2.4; policy embedded-offline-reviewed-trust-v3.
Evidence 3	Offline embedded-manifest verification; no network trust-list, manifest or revocation refresh.
Evidence 4	No reviewed local C2PA trust snapshot installed. Provenance overrides are disabled.

Provider evidence / 5

Field	Recorded detail
Provider	Local forensic cues
Model	Not recorded
Status	ok
Assessment	partially synthetic
Confidence	low
Role in assessment	Additional finding
Provider summary	Heuristic synthetic likelihood 41/100 from metadata + pixel statistics.
Evidence 1	No camera EXIF (common in AI exports, screenshots, and re-saves).
Evidence 2	ELA mean 1.8, peak 62 (uniform high ELA can indicate generation/heavy edit).
Evidence 3	High-frequency noise std 11.38 (very low/very uniform noise is a synthesis cue).
Evidence 4	Spectral peakiness 19.81 (periodic frequency peaks can mark generated textures).

Evidence supporting this verdict

Field	Recorded detail
Claude Vision	synthetic likelihood 80/100; tells: uniform soft airbrushed painterly shading across all elements, typical of current image-generation models, whole composite (portrait, anatomy, inset, icons, typography) rendered in one seamless style rather than assembled from separate assets, anatomically loose vasculature: arteries drawn over the brain surface and an idealized carotid path from neck to cortex, generic glowing highlights (red cortical hotspot, yellow injection-site glow) used as stylized emphasis
Gemini Vision	synthetic likelihood 68/100; tells: model read this as AI editing over a photographic base image, sterile digital rendering style, slightly inconsistent anatomical blending in syringe and hand, typical AI-generated vector/digital medical illustration texture

Additional findings

Field	Recorded detail
OpenAI Vision	synthetic likelihood 45/100; tells: Entire image is a medical illustration, with no camera-photo base visible., Headings and labels are legible, with consistent serif typography and aligned layout., Anatomy and cellular elements use smooth shaded fills and clean outlines compatible with conventional digital illustration., Repeated particle motifs have variable internal squiggles, but no decisive generation artifact is visible.
Local forensic cues	Supporting observation; considered alongside the model assessments.
C2PA Content Credentials	Embedded credentials: none found. Origin remains unresolved by this check.

Visible text

Field	Recorded detail
Recorded text	Biomedical Signals • Research Commentary • 2025 | mRNA Vaccination and the Brain: A Proposed Pathway | CAN PARTICLES CROSS THE BLOOD–BRAIN BARRIER? | Brain | Bloodstream | Suggested outcomes | Inflammation | Brain fog | Memory changes | Figure 1. Hypothetical pathway illustrated for discussion.

How this assessment was reached

Field	Recorded detail
1. override	outcome=none; detail=No confirmed C2PA AI assertion or SynthID watermark. Origin remains unresolved by these checks.
2. model_agreement	synthetic=['Claude Vision', 'Gemini Vision']; authentic=[]; uncertain=['OpenAI Vision']; forensic_supports_synthetic=False
3. verdict	value=synthetic_likely; confidence=medium; reason=2 models agree this is likely synthetic.

File and analysis record

Field	Recorded detail
File	journal-diagram.png
Analysis ID	8faa3dd9-dfff-476f-a33e-cd661e1e6f35
Generated	2026-09-26T22:31:05+00:00
Tool	SDA Vision 0.1.0
Models	claude-opus-5-5, gemini-3.8-flash, gpt-6-astra
Method version	pdf-preparation-2026-09-26.1
Analysis lens	general
Original SHA-256	c9a680547869618d7c63648d317b1f27c03ede59d5113a7cd6818f5ce2bd654e
Ingestion note 1	Fresh provider replies acquired for eleven-examples-refresh-20260926-r1; report assembled locally from the saved responses without further API calls.
Ingestion note 2	Provider latency values are restored from the acquisition receipts (maximum sampled-frame latency per provider); raw replies and assessment results are unchanged by this bookkeeping step.